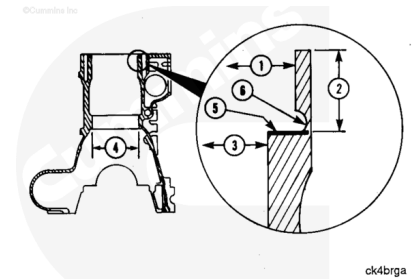


Trainings ▾

General Information

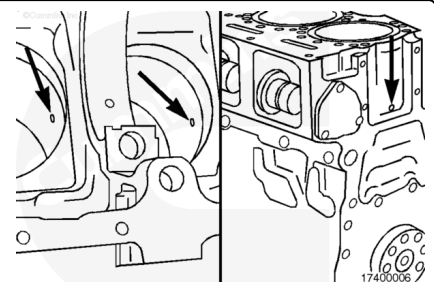
Definition of numerical terms (call-outs):

1. Upper counterbore inside diameter
2. Counterbore depth
3. Lower counterbore inside diameter
4. Packing ring bore
5. Counterbore ledge
6. Counterbore radius.



Rail Applications

- To allow for the proper coolant drainage of the cylinder head and engine block on a horizontally mounted QSK19 Rail engine, a coolant drain drilling was added to the cylinder block. The drilling runs the entire length of the cylinder block and passes through the center of each cylinder liner cavity. The drilling is plugged by a 1/4-inch pipe plug.
- Do **not** remove the plug.



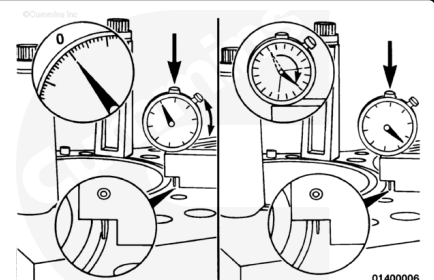
Inspect for Reuse

Measure the liner protrusion.

If the protrusion is **not** correct, remove the liner. The protrusion can be adjusted using seal rings and machining the counterbore ledge.

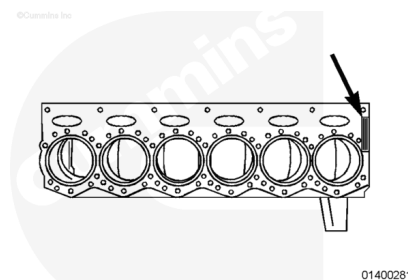
Refer to Procedure 001-064 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-064-tr.html)



Inspect the cylinder block to determine if an oversize or a standard size head gasket is required.

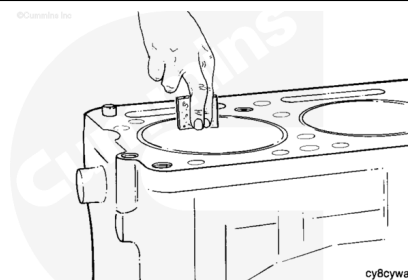
Note : An oversize head gasket may be required on blocks that have had material removed from the top surface of the block. Check the block for markings indicating an oversized gasket is required.



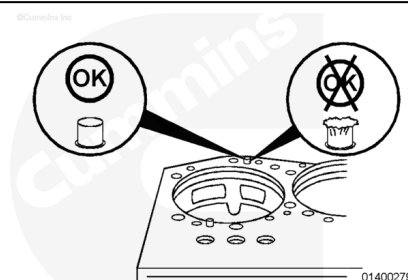
Use a scraper or a fibrous abrasive pad such as Scotch-Brite™ 7477, or equivalent, and solvent. Clean the top deck surface of the block. Do **not** allow any dirt into the cylinder. The surface **must** be free of dirt, oil, and gasket material, but does **not** have to look like new metal.

Note : Check the top surface of the block for wear. If fretting damage is present in an area where a head gasket seal ring or a grommet makes contact, the surface **must** be repaired.

Note : Fretting damage in any other area is acceptable **only** if it does **not** change the protrusion measurement of the counterbore or liner.



Inspect the groove pins for damage. Use the dowel pin extractor, Part Number ST-1134 , or equivalent, to remove the pins.

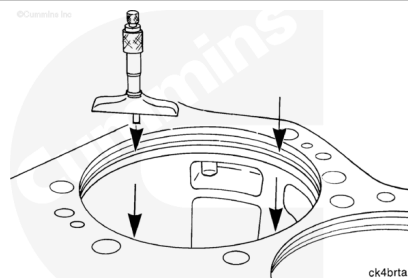


Measure

Note : Be sure the micrometer contacts the flat surface of the ledge. It **must not** touch the radius.

Note : If the counterbore depth does not meet specifications, a thicker or thinner seal ring may have been used in a previous machining bore process. Measure and record the measurement so that the liner protrusion can be determined.

Use a depth micrometer and measure the counterbore depth at the four places shown.

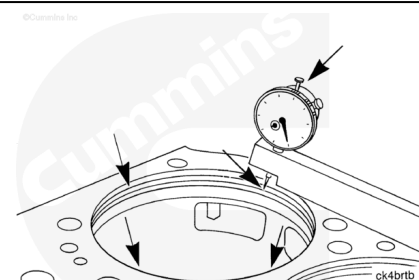


Counterbore Depth		
mm		in
13.755	MIN	0.5415
13.805	MAX	0.5435

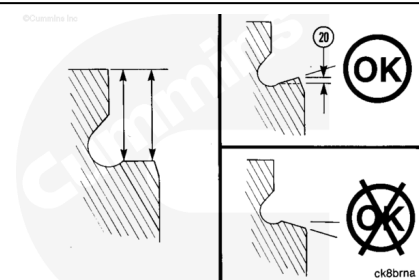
The four measurements **must not** vary more than 0.25 mm [0.001 inch]. If the measurements exceed the specifications, the counterbore ledge **must** be machined.

Note : Be sure that the indicator does **not** contact the counterbore radius on a block that does **not** have a double undercut.

Use depth gauge assembly, Part Number 3823495. Measure the angle of the counterbore ledge at four places on the counterbore circumference. The measurement of the ledge depth **must** be as near to the counterbore radius as possible, and as near to the counterbore edge as possible.



The angle (20) of the counterbore ledge is acceptable (OK) if the measurement that is near the counterbore edge is the same or no more than 0.036 mm [0.0014 inch] shorter than the measurement near the counterbore radius. If the measurement near the counterbore ledge is longer than the measurement near the counterbore radius, the ledge **must** be machined.



Measure the inside diameter of the upper counterbore (1). The point of measurement **must** be within 2.54 mm [0.100 inch] from the top of the block.

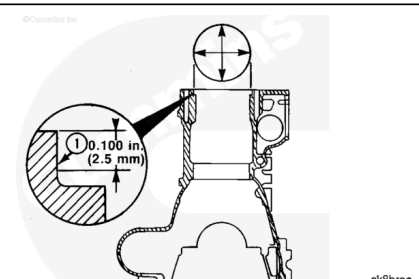
The inside diameter of the upper counterbore **must** be completely round within 0.076 mm [0.003 inch]. If the measurement is **not** within specification, check to see if the block can be machined to use oversize liners.

Use the following procedure for available oversized cylinder liners. Refer to Procedure 001-028 in Section 1.

(/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)

Note : The upper counterbore **must** be no more than 0.076 mm [0.003 inch] larger than the cylinder liner flange.

Note : If the counterbore does **not** meet specifications, an oversize liner may have been used in a previous



machining bore process. Measure and record the measurement so that the proper press fit liner can be determined.

**Counterbore Diameter - Upper Press Fit Diameter
Mechanically Actuated Injectors**

mm	Oversize [inch]	in
188.164 to 188.214	Standard	7.408 to 7.410
188.672 to 188.722	.020	7.428 to 7.430
189.180 to 189.230	.040	7.448 to 7.450
189.688 to 189.738	.060	7.468 to 7.470
190.272 to 190.322	.083	7.491 to 7.193
190.298 to 190.348	.084	7.492 to 7.494
190.577 to 190.627	.095	7.503 to 7.505

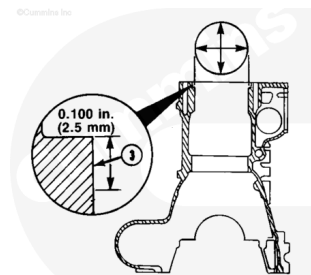
**Counterbore Diameter - Upper Press Fit Diameter
Electronically Actuated Injectors**

mm	Oversize [inch]	in
188.164 to 188.214	Standard	7.408 to 7.410
188.418 to 188.468	.010	7.418 to 7.420
188.672 to 188.722	.020	7.428 to 7.430
188.977 to 189.027	.032	7.440 to 7.442
189.180 to 189.230	.040	7.448 to 7.450

Measure the inside diameter of the lower counterbore (3). The point of measurement **must** be within 2.54 mm [0.100 inch] from the top of the counterbore ledge.

The inside diameter of the lower counterbore **must** be completely round within 0.050 mm [0.002 inch].

Note : If the counterbore does not meet specifications, an oversize liner may have been used in a previous



ck4brna

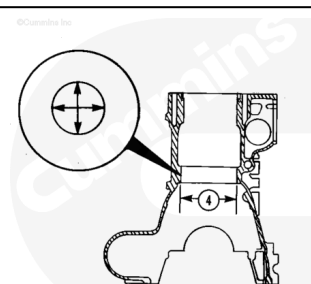
machining bore process. Measure and record the measurement so that the proper press fit liner protrusion can be determined.

Counterbore Diameter - Lower Press Fit Diameter Mechanically Actuated Injectors		
mm	Oversize [inch]	in
180.086 to 180.136	Standard	7.090 to 7.092
180.594 to 180.644	.020	7.110 to 7.112
181.737 to 181.787	.065	7.155 to 7.157
Counterbore Diameter - Lower Press Fit Diameter Electronically Actuated Injectors		
mm	Oversize [inch]	in
180.086 to 180.136	Standard	7.090 to 7.092
180.340 to 180.390	.010	7.100 to 7.102
180.594 to 180.644	.020	7.110 to 7.112
180.848 to 180.898	.030	7.120 to 7.122
181.102 to 181.152	.040	7.130 to 7.132

Check the chamfer at the top of the packing ring bore.
Excessive pitting **must** be repaired. See the Alternative Repair Manual, Bulletin 3379035.

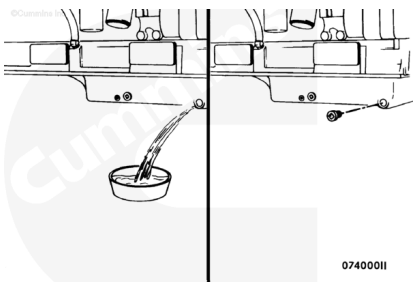
(4) Packing Ring Bore

mm		in
177.34	MIN	6.982
177.40	MAX	6.984

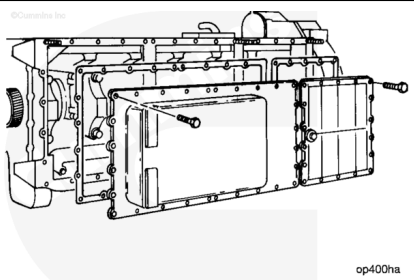


Leak Test

Drain the engine lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)

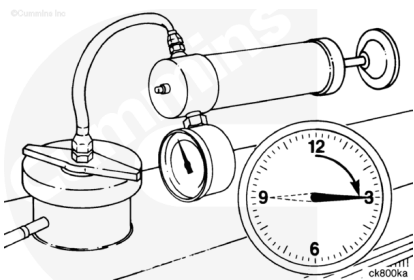


Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html)



Pressurize the engine cooling system. Refer to Procedure 018-018 in Section 18. (/qs3/pubsys2/xml/en/procedures/20/20-018-018.html)

Apply the air pressure 15 minutes before inspecting the cylinder liner seats. Be sure the system is holding air pressure before beginning inspection.



Measurements

	kpa	psi
Air Pressure	138	20

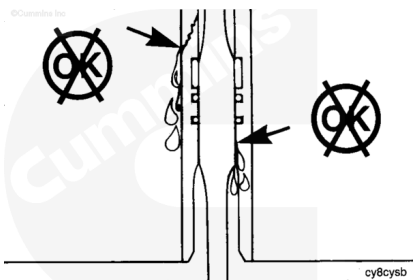
Inspect the outside diameter of the cylinder liners and the area below the cylinder liner seats in the cylinder block for coolant leaks.

If a leak is found:

- Remove the cylinder liner(s). Inspect the sealing ring and cylinder liner. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html)

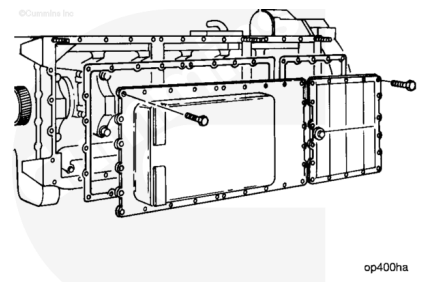
If a leak is found in the cylinder liner seat:

Remove the cylinder liner(s). Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-028-tr.html) Inspect the cylinder block liner counterbore area.



Install a new gasket and the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-025-tr.html>)

Torque Value: 88 n•m [65 ft-lb]



Fill the engine with clean 15W-40 oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/20/20-007-037.html>)

Operate the engine and check for coolant or lubricating oil leaks.

